UNITEDWORLD INSTITUTE OF TECHNOLOGY (UIT)

B.TECH. COMPUTER SCIENCE AND ENGINEERING (BTCE)

Vision

The vision of Unitedworld School of Computational Intelligence (USCI) is to be known globally as a School of Excellence that provides transformative educational experience, creating positive societal impact through establishment of global centers of higher learning in emerging technology areas of computational intelligence in pursuit of academic excellence.

Mission

- To foster critical thinking amongst students and instilling in them the values, skills and attitude to become lifelong learners and efficient problem solvers.
- To ensure seamless integration of academics, research and innovation for nurturing professional excellence and entrepreneurship.
- To create an environment for holistic development of learners by strengthening student centric welfare activities in pursuit of building socially responsible citizens.

Programme Educational Objectives (PEOs)

PEO1: To prepare graduates with a strong foundation in Computer Science and Applications and problem solving &programming skills in order to build successful careers professionals in industry, government, academia, research, entrepreneurial pursuit and consulting firms.

PEO2: To equip students with analytical, design, development and soft skill to find innovative solutions to the real-world problems in collaboration with industry and professional societies.

PEO3: To inculcate entrepreneurship, managerial skills and team work in our students through demonstration of good analytical, design and implementation skills for the betterment of individual and society at large.

PEO4: To produce graduates who are ethical, socially responsible and lifelong learners to fulfill their goals.

Programme outcomes (POs)

PO1: Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2: Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3: Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4: Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5: Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.

PO6: The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7: Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9: Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11: Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12: Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Programme Specifics Outcome (PSO) (Data Science)

PSO1: Ability to work with emerging technologies in computing.

PSO2: Demonstrate Engineering Practice learned through industry internship and research project to solve live problems in various domains.

DEPARTMENT OF COMPUTER ENGINEERING (DCE)
B. Tech. Computer Science and Engineering (Programme Code- BTCE)

CURRICULUM

S No	Course Type	Course Code	Course Title	L	T	P	C	Hrs. / Week	IA	ET	Tot.
SEMESTER I (July 2024-Nov 2024)											
1	ES11	24BTESCX11	Problem Solving & Programming	3	0	1	4	5	50	50	100
2	UWE	24KUCC102	Critical Thinking	1	0	1	2	3	50	50	100
3	UWE	24KUCC101	Fundamentals of AI	1	0	1	2	3	50	50	100
4	BS11	24BTBSCX11	Engineering Mathmatics - I	3	1	0	4	4	50	50	100
5	BS12	24BTBSCX12	Applied Physics	2	0	1	3	4	50	50	100
6	HS11	24BTHSCX11	Professional Communication (Training)	2	0	0	2	2	50	50	100
7	ES12	24BTESCX12	Basic Electrical and Electronics Engineering	2	0	1	3	4	50	50	100
8	PC11	24BTPCCX11	Unix and Shell Programming	1	0	1	2	3	50	50	100
			Total	**15**	**1**	**6**	**22**	**28**			
SEMESTER II (AIML) (Dec 2024-April 2025)											
9	PC21	24BTPCCX21	Python Programming	3	0	1	4	5	50	50	100
10	UWE		Psychology	1	0	1	2	3	50	50	100
11	PC22	24BTPCCX22	Data Structures	3	0	1	4	5	50	50	100
12	BS21	24BTBSCX21	Engineering Mathmatics - II	3	1	0	4	4	50	50	100
13	MC21	24BTMCCX21	Indian Knowledge system (SWAYAM-MOOC)	2	0	0	0	2	50	50	100
14	ES21	24BTESCX21	Engineering Design and Workshop Practice Simulation	2	0	1	3	4	50	50	100
15	UWE		Applied AI	1	0	1	2	3	50	50	100
16	VA21	24BTVACX21	Red Hat Aademy Course (GSS-RHA)	2	0	0	0	2	50	50	100
			Total	17	1	5	**19**	**28**			
SEMESTER III (July 2025-Nov 2025)											
17	PC31	24BTPCCX31	Design and Analysis of Algortihms	3	0	1	4	3	50	50	100
18	PC32	24BTPCCX32	Operating System	3	0	1	4	3	50	50	100
19	PC33	24BTPCCX33	Database Management Systems	3	0	1	4	3	50	50	100
20	BS31	24BTBSCX31	Discrete Mathematical Structures	3	0	0	3	3	50	50	100
21	ES31	24BTESCX31	Ethics in Engineering Practice (NPPTEL-MOOC)	2	0	0	2	-	50	50	100
22	PE3X		Programme Core/Specialization Elective - 1	4	0	0	4	3	50	50	100
23	HS31	24BTHSCX31	Entrepreneurship Development & IPR (Workshop)	2	0	0	2	3	50	50	100

S No	Course Type	Course Code	Course Title	L	T	P	C	Hrs. / Week	IA	ET	Tot.
24	VA31	24BTVACX31	Oracle Academy Course (GSS-OA)	2	0	0	0	4	50	50	100
			Total	**22**	**0**	**3**	**23**	**28**			
25	**PC299**	**24BTPC299**	**Summer Project (Application/Research) - 1**	0	0	3	3	6	50	50	100
SEMESTER IV (Dec 2025-April 2026)											
26	PC41	24BTPCCX41	Computer Architecture and Organization	3	1	0	4	4	50	50	100
27	PC42	24BTPCCX42	Object Oriented Programming using C++	2	0	2	4	6	50	50	100
28	BS41	24BTBSCX41	Probability and Statistics	3	0	0	3	3	50	50	100
29	HS41	24BTHSCX41	Environmental Science (NPTEL - MOOC)	2	0	0	0	2	50	50	100
30	PE4X		Programme Core/Specialization Elective - 2	4	0	0	4	4	50	50	100
31	PE4X		Programme Core/Specialization Elective - 3	4	0	0	4	4	50	50	100
32	VA41	24BTVACX41	EC-Council Academy Course (GSS-ECC)	2	0	0	0	2	50	50	100
			Total	20	1	2	**19**	**25**			
SEMESTER V (July 2026-Nov 2026)											
33	PC51	24BTPCCX51	Programming using Java	3	0	1	4	5	50	50	100
34	PC52	24BTPCCX52	R Programming	2	0	1	3	4	50	50	100
35	PC53	24BTPCCX53	Computer Networks	3	0	0	3	3	50	50	100
36	BS51	24BTBSCX51	Optimization Techniques	3	0	0	3	3	50	50	100
37	PE5X		Programme Core/Specialization Elective - 4	4	0	0	4	4	50	50	100
38	PE5X		Programme Core/Specialization Elective - 5	4	0	0	4	4	50	50	100
39	VA51	24BTVACX51	Personality Development (Activity Based Learning)	2	0	0	0	2	50	50	100
40	HS51	24BTHSCX51	Financial Literacy (Workshop)	2	0	0	2	2	50	50	100
			Total	23	0	2	**23**	**27**			
41	**PC399**	**24BTPC399**	**Summer Project (Application/Research) - 2**	0	0	6	**6**	**12**	50	50	100
SEMESTER VI (Dec 2026-April 2027)											
42	PC61	24BTPCCX61	Data Mining and Analysis	3	0	1	4	5	50	50	100
43	PC62	24BTPCCX62	Software Engineering	3	0	0	3	3	50	50	100
44	PC63	24BTPCCX63	Full Stack Development	0	0	2	2	4	50	50	100
45	HS61	24BTHSCX61	Community Service (Activity Based Learning - ABL)	0	0	1	1	2	50	50	100
46	PE6X		Programme Core/Specialization Elective - 6	4	0	0	4	4	50	50	100
47	PE6X		Programme Core/Specialization Elective - 7	4	0	0	4	4	50	50	100

S No	Course Type	Course Code	Course Title	L	T	P	C	Hrs. / Week	IA	ET	Tot.
48	BS61	24BTBSCX61	Research Methods in Computer Science (NPTEL-MOOC)	2	0	0	2	2	50	50	100
49	VA61	24BTVACX61	Quantitative Aptitude & Logical Reasoning (Training)	0	0	1	1	2	50	50	100
50	OE		Open Elective - University - 1	3	0	0	3	3	50	50	100
			Total	19	0	5	**24**	**29**			
SEMESTER VII (July 2027-Nov 2027)											
51	PC71	24BTPCCX71	Big Data Analytics (PBL)	3	0	1	4	5	50	50	100
52	PE7X		Programme Core/Specialization Elective - 8	4	0	0	4	4	50	50	100
53	PC72	24BTPCCX72	Application/Research Project - 3	0	0	7	7	14	50	50	100
			Total	7	0	8	**15**	**23**			
SEMESTER VIII (Dec 2027 - April 2028)											
54	PC81	24BTPCCX81	Semester Capstone Project (Industry/Research) - 4	0	0	10	10	20	50	50	100
			Total	0	0	10	**10**	**20**			

Important Note:
1. As per UGC/AICTE Regulation/Gazette Notifications, at least one subject must be studied on Approved MOOC Platform with credit transfer compulsorily. Maximum count of MOOC Courses with credit transfer shall be up to 40% of total credit.
2. Semester MOOC Course will be declared by the school just before the commencement of every semester and all students have to register and appear for the examinations conducted by the MOOC Platform without fail.
3. This MOOC course must be of 8 to 12 weeks duration and carries 3 or 4 Credits as specified in the curriculum.

Summary of Credits Distribution

Sem.	I	II	III	IV	V	VI	VII	VIII	Tot. Cr.
Credits	**22**	**19**	**23+3**	**19**	**23+6**	**24**	**15**	**10**	**164**
Cum. Cr.	**-**	**41**	**67**	**86**	**115**	**139**	**154**	**164**	

Electives Basket

(Students will select one elective course from each basket) (4 Credits)

Elective Baskets	B.Tech. CSE	
	Course Code	**Course Name**
Elective 1	24BTPEEX31	Introduction to Data Science
	24BTPECS32	Fundamentals of IoT
Elective 2	24BTPEEX41	Cloud Computing
	24BTPECS42	Introduction to Artificial Intelligence
Elective 3	24BTPECS43	Applications of Machine Learning
	24BTPEEX44	Data Engineering
Elective 4	24BTPECS51	Introduction to FinTech
	24BTPECS52	Ethical Hacking
Elective 5	24BTPECX53	Digital Image Processing
	24BTPECX54	Introduction to Generative AI
Elective 6	24BTPECS61	Introduction to Blockchain
	24BTPECS62	Fundamentals of Deep Learning
Elective 7	24BTPECS63	Applied Cryptography
	24BTPECS64	Predictive Analytics
Elective 8	24BTPECX71	Recommender Systems
	24BTPECS72	Human Computer Interaction

SEMESTER I

COURSE			Credit Structure & Total Hours					Evaluation Scheme		
SNo	**Course Code**	**Course Title**	**L**	**T**	**P**	**C**	**Hrs. / Week**	**IA**	**ET**	**Tot.**
1	24BTESCX11	Problem Solving & Programming	3	0	1	4	5	50	50	100
2	24KUCC102	Critical Thinking	1	0	1	2	3	50	50	100
3	24KUCC101	Fundamentals of AI	1	0	1	2	3	50	50	100
4	24BTBSCX11	Engineering Mathematics -I	3	1	0	4	4	50	50	100
5	24BTBSCX12	Applied Physics	2	0	1	3	4	50	50	100
6	24BTHSCX11	Professional Communication (Training)	2	0	0	2	2	50	50	100
7	24BTESCX11	Basic Electrical and Electronics Engineering	2	0	1	3	4	50	50	100
8	24BTPCCX11	Unix and Shell Programming	1	0	1	2	3	50	50	100
		Total	15	2	6	**22**	**28**			

Course code	Course Name	Hours/week			Credit	Max. Marks
		L	T	P	C	
24BTESCX11	Problem Solving & Programming	3	0	1	4	100
Pre-requisite	NA					

Evaluation Scheme	Theory	Hours	Marks
	End Semester Examination	2-2.5	50
	Internal Assessment	-	50
	1. Mid Semester Examination (Theory)/Continuous Assessment (Practical)	1.5	30
	2. Attendance	-	10
	3. Assignment/Seminar/Activity/ School Approved Certificate Course/Events/Professional Societies Membership and Participants	-	10

UNIT-I	OVERVIEW OF C	9 Hours

How to develop a program - Algorithms - Flow-charts - Types of Programming Languages – Compiler, Linker, Loader - Testing and Debugging a program- Importance of C - Basic structure of C programs – Constants, Variables and Data Types - Character set - C Tokens, Keywords and Identifiers - Declaration of variables - Assigning values to variables- Defining symbolic constants. Operators and expression: Types of Operators - Operator precedence and associativity-Arithmetic Expressions - Evaluation of expressions - Type conversions in expressions.

UNIT-II	DECISION MAKING AND BRANCHING	9 Hours

Simple IF, IF-ELSE, Nested IF-ELSE, ELSE-IF ladder, Switch statement - GOTO statement - WHILE loop - DO WHILE loop - FOR loop - Arrays: Definition & Declaration - One dimensional - Two dimensional - Multi dimensional arrays - Dynamic arrays.

UNIT-III	STRINGS AND FUNCTIONS	9 Hours

Introduction - Declaring and initializing string variables - Reading strings from terminal - Writing strings to screen - String handling functions - Table of strings. Definition of functions - Return values and their types - Function calls - Function declaration - User Defined functions: Introduction and Need - A Multi- function program - Elements of user-defined function - Nesting of functions - Recursion - Passing arrays to functions - Passing strings to function.

UNIT-IV	STRUCTURES AND UNIONS	9 Hours

Introduction - Defining a structure - Declaring structure variables - Accessing structure members - Structure initialization - Copying and comparing structure variables - Arrays of structures - Arrays within structures - Structures within structures - Structures and functions - Unions - Pointers: Introduction - Understanding pointers - Accessing the address of a variable - Initializing of pointer variables. Chain of pointers
-Pointers and arrays - Arrays of pointers - Pointers as function arguments - Functions returning pointers - Pointers to functions.

UNIT-V	I/O AND FILE MANAGEMENT	9 Hours

Introduction - Managing input and output operations: Reading and writing a character - Formatted input and output, Defining and opening a file -Closing a file - Input/output operation on files - Error handling during I/O operations - Random access files - Command line arguments.

Total hours	45 periods

Course Outcome(s)	At the end of the course, the students will be able to:	
COs	**Statements**	**Bloom's Level**
CO1	Understand C programs using operators, type conversion and input-output functions.	**L2**

CO2	Apply decision making and looping statements using C programs.	**L3**
CO3	Develop C programs using the concepts of Arrays and strings.	**L3**
CO4	Analyse the usage of structures, unions and files in C.	**L4**
CO5	Apply the concepts of file Management using C programs.	**L3**

TEXT BOOK:

1.	E. Balagurusamy, “Programming with ANSI-C”, Eighth Edition, Tata McGraw Hill, 2019
2.	Hanly J R &Koffman E.B, “Problem Solving and Programm design in C”, PearsonEducation, 2009.

REFERENCES:

4.	Herbert Schildt, "C: The complete Reference", 4th Edition, Mc Graw Hill,2003.
5.	B.L. Juneja, “Programming in C", 1st Edition, Cengage Learning, 2012.

LIST OF EXPERIMENTS
1. Write a program to compute the addition, subtraction, product, quotient and remainder of two given numbers
2. Write a program to convert temperature from degree centigrade to Fahrenheit. °F = °C*9/5+32
3. Write a program to swap the values of two variables.
4. Write a program to print the odd and even numbers within a given range.
5. Write a program to find the factorial of a given number.
6. Write a program to print the sum of digit of a given number.
7. Write program to print the Fibonacci series up to a given number.
8. Write a program to print the prime numbers within a given range.
9. Write a program to determine and print the sum of the following harmonic series for a given value of n1 +1/2+1/3+1/4+……+1/n.
10. Write a program to print the following pattern (Take number of lines as input from the user). **** *** ** *
11. Write a C program to check if string is palindrome or not.
12.Write a program to read three integer values from the keyboard and display the output stating that they are the sides of right-angled triangle.
13.

(a) Write a program for addition of two nxm matrices. (b) Write a program for multiplication of two nxm matrics
14. Write a program to find out the length of a given string without using the library function strlen ().
15. a) Write a program to swap value of two variables using pointer. b)Write a program to accept student details (regNo, name, address, phoneNo) and store them in a file and perform the following operations on it. i). Search ii). Add iii). Delete iv). Modify v). Display.
Total hours : 30 periods

Course Outcomes with Program Outcomes

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11	PO 12
CO 1	2	1	-	2	-	3	1	-	-	-	-	3
CO 2	3	3	-	2	-	3	3	-	-	2	-	3
CO 3	3	3	-	2	-	3	3	-	-	2	-	3
CO 4	3	3	-	2	-	3	3	-	-	2	-	3
CO 5	2	2	-	2	-	3	1	-	-	2	-	2

Course code	Course Name	Hours/week			Credit	Max. Marks
24KUCC102	Critical Thinking	L	T	P	C	100
		1	0	1	2	
Pre-requisite	NA					

Evaluation Scheme	Theory	Hours	Marks
	End Semester Examination	2 - 2.5 Hr	50
	Internal Assessment	-	50
	1. Continuous Assessment (Tutorial)	1.5	30
	2. Attendance	-	10
	3. Assignment/Seminar/Activity/ School Approved Certificate Course/Events/Professional Societies Membership and Participants	-	10

UNIT-I	INTRODUCTION TO CRITICAL THINKING	3
Overview of Critical Thinking: Critical thinking vs. creative thinking, Elements of Critical Thinking: Observation and interpretation, Analysis and inference, Evaluation and explanation, Self-regulation, Benefits of Critical Thinking in Engineering: Problem-solving, Decision-making, Innovation.		
UNIT-II	LOGICAL REASONING AND ARGUMENTATION	3
Basics of Logical Reasoning: Deductive vs. inductive reasoning, Validity, soundness, and truth, Types of Arguments: Simple and complex arguments, Causal arguments, Analyzing Arguments: Identifying premises and conclusions, Evaluating evidence and assumptions, Common logical fallacies, Constructing Arguments: Structure and components of a strong argument, Persuasive techniques and refutation.		
UNIT-III	PROBLEM-SOLVING AND DECISION-MAKING	3
Problem-Solving Techniques: Identifying and analyzing the problem, Generating potential solutions, Evaluating and selecting the best solution, Decision-Making Models: Rational decision-making model, Bounded rationality, Intuitive decision-making model, Tools for Problem-Solving: Brainstorming and root cause analysis, SWOT analysis and decision trees, Flowcharts.		
UNIT-IV	CRITICAL THINKING IN COMPUTER ENGINEERING	3
Application of Critical Thinking in Software Development: Requirement analysis and system design, Code review and testing, Algorithmic Thinking: Understanding and designing algorithms, Analyzing algorithm complexity, Debugging and Troubleshooting: Identifying bugs and errors, Systematic debugging approaches, Tools and techniques for troubleshooting, Ethical and Social Implications: Ethical decision-making in computing, Social responsibility of computer engineers.		
UNIT-V	COMMUNICATION AND PRESENTATION OF CRITICAL THINKING	3
Effective Communication of Ideas: Clear and concise writing, Technical report writing, Presentation skills, Collaboration and Critical Thinking: Team-based problem-solving, Collaborative software development, Peer review and feedback, Presenting Critical Analysis: Structuring presentations, Visual aids and tools, Engaging the audience.		
	Total hours	15 Periods

Course Outcome(s)	At the end of the course, the students will be able to:	
COs	Statements	Bloom's Level
CO1	Develop a deep understanding of critical thinking principles and their application in engineering.	L1
CO2	Enhance logical reasoning and argumentation skills.	L2
CO3	Improve problem-solving and decision-making capabilities.	L3
CO4	Apply critical thinking to computer engineering tasks and projects.	L5
CO5	Communicate and present ideas effectively.	L6

TEXT BOOK:

1.	“Critical Thinking: Tools for Taking Charge of Your Professional and Personal Life" by Richard Paul and Linda Elder
2.	"A Rulebook for Arguments" by Anthony Weston
3.	"The Thinker's Toolkit: 14 Powerful Techniques for Problem Solving" by Morgan D. Jones
4.	"Introduction to the Theory of Computation" by Michael Sipser
5.	"The Elements of Style" by William Strunk Jr. and E.B. White
REFERENCES:	
1.	"Critical Thinking: Tools for Taking Charge of Your Professional and Personal Life" by Richard Paul and Linda Elder
2.	"The Thinker's Toolkit: 14 Powerful Techniques for Problem Solving" by Morgan D. Jones
3.	"A Rulebook for Arguments" by Anthony Weston

Note : Practical list will be provided by the subject handling faculty.

Course Outcomes with Program Outcomes

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**
CO1	3	3	1	1	1	1	1	3	-	-	-	3
CO2	3	3	2	2	2	-	-	1	-	-	-	3
CO3	3	3	3	3	3	-	-	-	-	-	2	2
CO4	3	3	3	3	3	3	3	3	2	2	2	3
CO5	-	-	2	2	2	-	-	-	3	3	3	2

Course code	Course Name	Hours/week			Credit	Max. Marks
24KUCC101	Fundamentals of AI	L	T	P	C	100
		1	0	1	2	
Pre-requisite	NA					

Evaluation Scheme	Theory	Hours	Marks
	Term End Summative Assessment: Case Studies/ PBL/Practical	-	50
	Internal Assessment	-	50
	1) Mid Semester Examination (Theory)/Continuous Assessment (Practical)	1.5	30
	2) Attendance	-	10
	3) Assignment/Seminar/Activity/ School Approved Certificate Course/Events/Professional Societies Membership and Participants	-	10

UNIT-I	INTRODUCTION TO AI	3
What is AI? History and evolution of AI.Types of AI: Machine Learning, Deep Learning, Symbolic AI. Key concepts: Algorithms, Data, Models, Learning. Science fiction vs. reality of AI.		
UNIT-II	**PROBLEM SOLVING WITH AI**	**3**
Search algorithms: Breadth-first search, Depth-first search. Knowledge representation and reasoning. Introduction to Machine Learning: Supervised vs. Unsupervised Learning.		
UNIT-III	**MACHINE LEARNING FUNDAMENTALS**	**3**
Machine Learning algorithms: Linear Regression, Decision Trees. Training and evaluating machine learning models. Hands-on Activity: Use a user-friendly platform (e.g., Google Teachable Machine, TensorFlow Playground) to build a simple image classification model.		
UNIT-IV	**APPLICATIONS OF AI IN YOUR DISCIPLINE**	**3**
How AI is impacting various fields: Law: Legal research, automated document review, predictive analytics for legal outcomes. Management: Business analytics, customer relationship management, intelligent automation. IT: Network security, software development, IT infrastructure management. Design: Design automation, personalized design experiences, AI-powered creativity tools. Dentistry: Medical imaging analysis, patient diagnosis support systems, robotic dentistry. Arts & Humanities: Algorithmic art creation, sentiment analysis of literature, music generation.		
UNIT-V	**THE FUTURE OF AI**	**3**
Ethical considerations of AI development and use (bias, fairness, transparency). Societal impact of AI: Jobs, privacy, decision-making. The future of AI and its potential benefits and challenges for various discipline.		
	Total hours	**15 periods**

Course Outcomes: At the end of the course, the students will be able to:

COs	Statements	Bloom's Level
CO1	Define AI and its core concepts, differentiating between types of AI	**L1**
CO2	Explain the working principles of AI, including fundamental algorithms and techniques.	**L3**
CO3	Evaluate the capabilities and limitations of AI systems, including ethical considerations.	**L3**
CO4	Identify real-world applications of AI relevant to various disciplines.	**L5**
CO5	Engage with basic AI tools and techniques through hands-on activities	**L2**

TEXT BOOK:

1. Sailaja Krishnamacharya and Mukesh Zaveri, "Artificial Intelligence for Everyone: Dive into the World of AI with Python", Practical and beginner-oriented.

Note : Practical list will be provided by the subject handling faculty.

Course code	Course Name	Hours/week			Credit	Max. Marks
		L	T	P	C	
24BTBSCX11	Engineering Mathematics-I	3	1	0	4	100
Pre-requisite	NA					

Evaluation Scheme	Theory	Hours	Marks
	End Semester Examination	2-2.5	50
	Internal Assessment	-	50
	1. Mid Semester Examination (Theory)/Continuous Assessment (Practical)	1.5	30
	2. Attendance	-	10
	3. Assignment/Seminar/Activity/ School Approved Certificate Course/Events/Professional Societies Membership and Participants	-	10

UNIT-I	Linear Algebra	9

Vector spaces, Subspaces, basis and dimension, linear transformations, representation of transformations by Matrices, linear functionals, transpose of linear transformations, canonical forms. Linear functionals and adjoints, Bilinear forms, symmetric bilinear forms, skew symmetric bilinear forms.

UNIT-II	Calculus	9

Continuity and differentiability of a function of single variable, statement of Rolle's Theorem, Lagrange's mean value theorem and applications. Double and Triple Integrals: Calculations, Areas, Volumes, change of variables.

UNIT-III	Vector Calculus	9

Applications. Integrals of Vector Functions: Line integrals, Green's formula, path independence, Surface integral: definition, evaluation, Stoke's formula, Gauss-Ostrogradsky divergence theorem.

UNIT-IV	Differential Equations	9

Ordinary Differential Equations: First order linear equations, Bernoulli's equations, Exact equations and integrating factor, Second order and Higher order linear differential equations with constant coefficients.

UNIT-V	Multivariate Calculus	9

Integral Calculus: Definite Integrals as a limit of sums, Applications of integration to area, volume, surface area, Improper integrals. Functions of several variables: Continuity and differentiability, mixed partial derivatives, local maxima and minima for function of two variables, Lagrange multipliers.

Total hours	60 periods

Course Outcomes: After completion of course, students would be able to:

COs	Statements	Bloom's Level
CO1	Understand basic linear algebra.	L2
CO2	Understand and apply calculus.	L3
CO3	Understand and apply vector calculus.	L3
CO4	Understand and apply differential equations.	L3
CO5	Understand and apply multivariate calculus.	L3

TEXT BOOKS/ REFERENCES	
1.	G. B. Thomas, R. L. Finney. Calculus and Analytic Geometry, Ninth Edition, Pearson Education, 2010
2.	Reena Garg, Advanced Engineering Mathematics, Khanna Book Publishing Co., Delhi.
3.	B. V. Ramana. Higher Engineering Mathematics, Tata McGraw Hill, 2017
4.	E. Kreyszig. Advanced Engineering Mathematics, Wiley, 2015
5.	Calculus and Analytic Geometry, G. B. Thomas and R. L. Finney, Pearson Education, 2010
MOOC BASED REFERENCES	
1	Basic calculus for Engineers, Scientists and Economists, Prof. Joydeep Dutta, IIT Kanpur on NPTEL/SWAYAM

Course Outcomes with Program Outcomes

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	3	1	1	-	-	-	-	-	-	-
CO2	3	3	3	-	-	-	-	-	-	-	-	-
CO3	3	3	2	-	-	-	-	-	-	-	-	-
CO4	3	2	3	-	-	-	-	-	-	-	-	-
CO5	3	3	3	-	-	-	-	-	-	-	-	-

Course code	Course Name	Hours/week			Credit	Max. Marks
		L	T	P	C	
24BTBSCXX2	Applied Physics	2	0	1	3	100
Pre-requisite	NA					

Evaluation Scheme	Theory	Hours	Marks
	End Semester Examination	2	50
	Internal Assessment	-	50
	1. Mid Semester Examination (Theory)/Continuous Assessment (Practical)	1.5	25
	2. Attendance	-	05
	3. Assignment/Seminar/Activity/ School Approved Certificate Course/Events/Professional Societies Membership and Participants	-	20

UNIT-I	**ULTRASONIC WAVES & NON-LINEAR OPTICS**	**6**

Introduction, Properties, Production method of Ultrasonic Waves, Velocity measurement using acoustic grating, SONAR, Application of Ultrasonic Waves, LASER:-Absorption and Emission, Einstein Coefficient Relation, Terms Used in LASER, Types of LASER, Ruby Laser, Application of LASER, Fibre Optics-Construction, Principle, Terms Used in Fibre Optics and Classification of Optical Fibre.

UNIT-II	**DIELECTRIC MATERIALS**	**6**

Basic terminology, Classification based on Dipole Moment, Relation between Permittivity, Susceptibility and Polarization, Types of Polarization, Langevin-Debye Equation, Definition and Formula of Internal Field, Clausius-Mossotti Equation, Types of Dielectric Materials, Application of Dielectric Materials.

UNIT-III	**PROPERTIES OF ELECTRONIC MATERIALS**	**6**

Basic terminology, Types of electronic materials, Classical free electron theory, Expression for electrical conductivity, Thermal conductivity expression, Wiedemann-Franz Law, Success and Failures of Classical Free Electron Theory, Quantum Free Electron Theory, Success and Failures of Quantum Free Electron Theory, Fermi-Dirac statistics.

UNIT-IV	**SEMICONDUCTOR MATERIALS**	**6**

Introduction, Properties of Semiconductor, Types of Semiconductor, Carrier concentration: Density of electrons in the conduction band and density of holes in the valence band, Fermi level and its variation with temperature for Intrinsic Semiconductor, Mobility and conductivity, Velocity-electric field relations, Drift and diffusion transport - Einstein ‘s relation, Hall effect, and experimental verification.

UNIT-V	**SUPERCONDUCTIVITY**	**6**

Introduction, Properties of Superconductor, Meissner Effect, Pressure Effect, Effect of Magnetic Field, Impurity Effect, Isotopic Mass Effect, Mechanism of Superconductivity: BCS Theory, Penetration Depth: Magnetic Field, Josephson’s Junction and Its Application, Application of Superconductor.

Total hours	**30 Periods**

Course Outcome(s)	At the end of the course, the students will be able to:	
COs	**Statements**	**Bloom’s Level**
CO1	Explain the mechanism of Ultrasonic Waves, types of LASER, and construction and working of Optical Fibre.	**L3**
CO2	Explain the properties and characteristics of dielectric materials.	**L2**
CO3	Explain the fundamentals of classical and quantum electron theories and energy band structures.	**L1**
CO4	Summarize the characteristics of semiconductors.	**L1**
CO5	Explain the phenomena of Superconductivity.	**L2**

TEXT BOOK:

1.	Dr.M. Arumugam, “Engineering Physics - Anuradha Agencies”, Kumbakonam- 3rd Revised Edition 2002.
2.	Umesh K Mishra & Jasprit Singh, “Semiconductor Device Physics and Design”, Springer,2008.
3.	Wahab, M.A. “Solid State Physics: Structure and Properties of Materials”, Narosa, Publishing House, 2009.
REFERENCES:	
4.	Rolf Enderlein & Norman J Horing, Fundamentals of Semiconductor Physics and Devices, Verlag, 2012.
5.	Kasap, S.O., “Principles of Electronic Materials and Devices”, McGraw-Hill Education, 2007.
6.	Kailash K. Sharma, “Optics: Principles and Applications”, kindle edition-2006.

Course Outcomes with Program Outcomes

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11	PO 12
CO 1	3	-	-	-	-	3	-	-	1	2	1	1
CO 2	2	-	-	-	-	3	-	-	1	2	1	1
CO 3	1	-	-	-	-	2	-	-	1	2	1	1
CO 4	1	-	-	-	-	2	-	-	1	2	1	1
CO 5	1	-	-	-	-	2	-	-	1	2	1	1

Course Outcomes	LIST OF EXPERIMENT	
CO 1	Determination of wavelength of mercury spectrum – spectrometer grating.	
	Determination of acceptance angle in an optical fiber.	
CO 2	Air Wedge – Thickness of a Wire.	
	Determination of Band Gap of a semiconductor material.	
	Determination of viscosity of liquid – Poiseuille‟s method.	
	Torsional Pendulum – Determination of Rigidity modulus 10.	
CO 3	Determination of velocity of sound and compressibility of liquid – Ultrasonic interferometer.	
CO 4	Determination of Young's modulus by Non uniform bending method.	
	Determination of Young's modulus by Uniform bending method.	
CO 5	Determination of Wavelength, and particle size using Laser.	
Course Outcome(s)	At the end of the course, the students will be able to:	
COs	**Statements**	**Bloom's Level**
CO 1	Apply the Principle of Optics to evaluate engineering properties of the material.	**L3**
CO 2	Apply the Principle of Thermal Physics to evaluate engineering properties of the material.	**L3**
CO 3	Explain the working principles and properties of optical properties of optical materials and gain knowledge about Ultrasonic and their applications in engineering fields.	**L4**
CO 4	To Study the Elastic Properties of the Materials for Uniform and Non Uniform Method.	**L4**
CO 5	To Study Light Properties of the Material using the Light Amplification Stimulated Emission (LASER) method.	**L4**
TEXT BOOK:		
1	Dr.M. Arumugam, "Engineering Physics - Anuradha Agencies", Kumbakonam- 3 rd Revised Edition 2002.	
2	Umesh K Mishra & Jasprit Singh, "Semiconductor Device Physics and Design", Springer,2008.	
REFERENCES :		
3	Rolf Enderlein & Norman J Horing, Fundamentals of Semiconductor Physics and Devices, Verlag, 2012.	
4	Hanson G.W., "Fundamentals of Nanoelectronics", Pearson Education, 2009.	

Course code	Course Name	Hours/week			Credit	Max. Marks
		L	T	P	C	
24BTHSCX11	Professional Communication (Training)	2	0	0	2	100
Pre-requisite	NA					

Evaluation Scheme	Practical	Hours	Marks
	End Semester Examination	-	50
	Internal Assessment	-	50
	1. Continuous Assessment (Practical)	1.5	30
	2. Attendance	-	10
	3. Assignment/Seminar/Activity/ School Approved Certificate Course/Events/Professional Societies Membership and Participants	-	10

UNIT-I	PROFESSIONAL COMMUNICATION SKILLS	5
Importance of effective communication in the professional world, Written and oral communication techniques, Active listening and effective questioning, Non-verbal communication skills.		
UNIT-II	EFFECTIVE TEAMWORK AND COLLABORATION	5
Understanding the value of teamwork in the workplace, Team dynamics and roles, Conflict resolution and negotiation skills, Delegation of roles and responsibilities, Effective collaboration using project management tools.		
UNIT-III	PROJECT MANAGEMENT SKILLS	10
Introduction to project management, Project Life Cycle Management, Project time and cost estimation, Resource Management and Cost Management, Project Evaluation.		
UNIT-IV	PROFESSIONAL ETHICS AND PROJECT DEVELOPMENT	10
Ethics in profession and decision making, Project Modules Development, Project progress and Performance Management.		
UNIT-V	PROJECT DEMONSTRATION, PRESENTATION AND DOCUMENTATION SKILLS	5
Demonstration of project, Presentation, Report Formulation and Documentation.		
	Total hours	30 Periods

Course Outcome(s)	At the end of the course, the students will be able to:	
COs	Statements	Bloom's Level
CO1	Demonstrate effective communication skills in professional contexts.	L1
CO2	Understand the importance of teamwork and collaborate effectively with peers.	L2
CO3	Apply project management techniques to analyse and manage projects.	L3
CO4	Comprehend professional ethics, ethical decision-making and project development in the field of computer science.	L5
CO5	Develop project modules, demonstration, professional documents and presentations for various purposes.	L6

TEXT BOOK:

1. "Business Communication: Building Critical Skills" by Kitty O. Locker and Stephen Kyo Kaczmarek
2. "Teamwork and Teamplay" by Sivasailam "Thiagi" Thiagarajan and Glenn Parker
3. "Project Management: A Systems Approach to Planning, Scheduling, and Controlling" by Harold Kerzner
4. "Ethics in Engineering" by Mike W. Martin and Roland Schinzinger
5. "The Art of Presentation: Your Competitive Edge" by James O'Rourke

REFERENCES:

1. "Effective Business Communication" by Herta A. Murphy, Herbert W. Hildebrandt, and Jane P. Thomas

2.	"Crucial Conversations: Tools for Talking When Stakes Are High" by Kerry Patterson, Joseph Grenny, Ron McMillan, and Al Switzler
3.	"Project Management for Dummies" by Stanley E. Portny
4.	"Professional Ethics and Human Values" by M. Govindarajan, S. Natarajan, and V.S. Senthil Kumar
5.	"The Craft of Scientific Writing" by Michael Alley

Course Outcomes with Program Outcomes

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11	PO 12
CO 1	2	1	2	-	-	-	2	1	2	-	-	2
CO 2	2	1	2	1	-	-	2	2	3	-	-	2
CO 3	2	1	2	1	-	-	2	1	2	-	3	2
CO 4	2	1	3	1	-	-	2	3	2	-	2	2
CO 5	2	1	2	1	-	-	2	1	2	-	-	2

Course code	Course Name	Hours/week			Credit	Max. Marks
		L	T	P	C	
24BTESCX11	Basic Electrical and Electronics Engineering	2	0	1	3	100
Pre-requisite	NA					

Evaluation Scheme	Theory	Hours	Marks
	End Semester Examination	2-2.5	50
	Internal Assessment	-	50
	1. Mid Semester Examination (Theory)/Continuous Assessment (Practical)	1.5	30
	2. Attendance	-	10
	3. Assignment/Seminar/Activity/ School Approved Certificate Course/Events/Professional Societies Membership and Participants	-	10

UNIT-I	BASIC ELECTRICAL	6

Basic Electrical circuit elements (R-L-C), Different Active & Passive component, Ohms Law Kirchhoff's current and voltage laws, Representation of different waveforms, combinations (series and parallel),– Introduction to Single phase and Three phase circuits-. Real power, reactive power, apparent power, power factor.

UNIT-II	TRANSFORMER AND ROTATING MACHINE	6

Construction and working Principle of Transformer, DC Generator, DC Motor, and Induction motor –emf equation, Different types of transformer & rotating machine, Applications.

UNIT-III	MEASUREMENTS AND INSTRUMENTATION	6

Classification of instruments – moving coil and moving iron meters – Induction type, dynamometer type wattmeter's– Energy meter – Measuring systems; Basic principles of measurement; Types of measurement; Cathode ray oscilloscope, Transducers.

UNIT-IV	SEMICONDUCTOR DEVICES AND APPLICATIONS	6

Types of material, Basic idea about forward bias, reverse bias of PN junction diode, Zener Effect and half wave rectifier, full wave rectifier, bridge rectifier, Zener Diode and its Characteristics , Bipolar Junction Transistor – BJT Configurations and its VI Characteristics- SCR- FET,MOSFET

UNIT-V	DIGITAL FUNDAMENTALS AND LOGIC CIRCUITS	6

Number System & Binary codes; Boolean Algebra & Logic gates; Combinational & Sequential logic circuits; A/D & D/Aconverter, de Morgan's theorem, Encoder

Total hours	30

Course Outcomes: At the end of the course, the students will be able to:

COs	Statements	Bloom's Level
CO1	Analyse electric circuits.	L4
CO2	Describe the working principles of electrical machines.	L2
CO3	Distinguish appropriate instruments for electrical measurement for a specific application.	L4
CO4	Apply the concepts of various electronic devices to perform simple applications.	L3

CO5	Recognize the concept of various digital fundamentals and conversions.	**L1**
TEXT BOOK:		
1.	V.K Mehta and Rohit Mehta, Principle of Electrical Engineering, S Chand & Company, 2008	
2.	J.B.Gupta Basic of Electrical Engineering	
3.	D. C. Kulshreshtha, "Basic Electrical Engineering", McGraw Hill, 2009	
REFERENCES:		
1.	B. L. Theraja, "Electrical Technology – Part I and II", S. Chand and Co. 2012	
2.	D. P. Kothari and I. J. Nagrath, "Basic Electrical Engineering", Tata McGraw Hill, 2010.	

Course Outcomes with Program Outcomes

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO 10**	**PO 11**	**PO 12**
CO 1	3	2	2	2	2	1	-	-	-	-	-	-
CO 2	2	2	3	3	2	2	-	-	-	-	-	-
CO 3	2	3	3	3	2	2	-	-	-	-	-	-
CO 4	3	3	3	3	2	3	-	-	-	-	-	-
CO 5	3	3	3	3	3	2	2	-	-	-	-	-

S.NO	**LIST OF EXPERIMENTS**
1.	Study of Electronic components and Equipment's – Resistor, color coding measurement of AC signal parameter (peak-peak, rms period,frequency) using CR
2.	Verification of Kirchhoff's voltage and current laws
3.	Verification of Ohm's law
4.	Study the functions of Function generator and CRO/DSO
5.	Half wave and full wave rectifiers with and without filters- Observethe waveforms on CRO.
6.	V-I Characteristics of diodes and Zener diodes
7.	V-I Characteristics of diodes and P N junction
8.	Study and verify logic gate
9.	Verification of Half & full adder
10.	Verification of Half & full substraction
11.	Design and implementation of a simple digital system (Mini Project)
	Total hours: 30 periods

Course code	Course Name	Hours/week			Credit	Max. Marks
		L	T	P	C	
24BTPCCX11	Unix and Shell Programming	1	0	1	2	100
Pre-requisite	NA					

Evaluation Scheme	Theory	Hours	Marks
	End Semester Examination	2-2.5	50
	Internal Assessment	-	50
	1. Mid Semester Examination (Theory)/Continuous Assessment (Practical)	1.5	30
	2. Attendance	-	10
	3. Assignment/Seminar/Activity/ School Approved Certificate Course/Events/Professional Societies Membership and Participants	-	10

UNIT-I	**INTRODUCTION**	**3**
Introduction to unix-Brief History-What is Unix-Unix Components-Using Unix-Commands in Unix-Some Basic Commands-Command Substitution-Giving Multiple Commands.		
UNIT-II	**UNIX FILE SYSTEM**	**3**
The File system –The Basics of Files-What's in a File-Directories and File Names-Permissions Nodes-The Directory Hierarchy, File Attributes and Permissions-The File Command knowing the File Type-The Chmod Command Changing File Permissions-The Chown Command Changing the Owner of a File-The Chgrp Command Changing the Group of a File.		
UNIT-III	**SHELL-COMMAND LINE STRUCTURE**	**3**
Using the Shell-Command Line Structure-Met Characters-Creating New Commands-Command Arguments and Parameters-Program Output as Arguments-Shell Variables- -More on I/O Redirection-Looping in Shell Programs.		
UNIT-IV	**FILTERS AND AWK**	**3**
Filters-The Grep Family-Other Filters-The Stream Editor Sed-The AWK Pattern Scanning and processing Language-Good Files and Good Filters.		
UNIT-V	**SHELL PROGRAMMING**	**3**
Shell Programming-Shell Variables-The Export Command-The Read Command-Positional Parameters -Branching Control Structures-Loop Control Structures-The Continue and Break Statement--The Expr Command		
	Total hours	**15 Periods**

Course Outcome(s)	At the end of the course, the students will be able to:	
COs	**Statements**	**Bloom's Level**
CO1	Understand the UNIX and its basic features	**L1**
CO2	Understand UNIX file system & Interpret UNIX Commands	**L1**
CO3	Understand Shell Command Structure and shell variables.	**L2**
CO4	Explain the working principles of Filters and AWK pattern Scanning	**L3**
CO5	Design and Develop Shell Programming.	**L2**

TEXT BOOK:

1.	Sumitabha Das: "UNIX – Concepts and Applications", (Chapters 1,2,4,6-9,11-14,17,19), Tata McGraw Hill, Noida, 4th Edition, 15th Reprint, 2011,
2.	Richard Blum, Christine Bresnahan: Linux Command Line and Shell Scripting Bible, 2ndEdition, Wiley,2014.
REFERENCES:	
3.	Behrouz A. Forouzan and Richard F. Gilberg: "UNIX and Shell programming", Cengage Learning, India,1st Edition, 2005.
5.	M G Venkatesh Murthy: "UNIX and Shell programming", Pearson Education, Delhi, 1st Edition, 2005.

Course Outcomes with Program Outcomes

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11	PO 12
CO 1	1	-	-	-	-	2	-	-	1	2	1	1
CO 2	1	-	-	-	-	2	-	-	1	2	1	1
CO 3	2	-	-	-	-	3	-	-	1	2	1	1
CO 4	3	-	-	-	-	3	-	-	1	2	1	1
CO 5	1	-	-	-	-	2	-	-	1	2	1	1

List of Experiments:

1.	Use of basic Unix Shell Commands: ls, mkdir, rmdir, cd, cat, banner, touch, file, wc, sort, cut, dd, dfspace, du, ulimit.
2.	Implement grep command.
3.	Implement commands related to inode, I/O redirection, piping, process control Commands, mail.
4.	Write and execute a simple script. Write a shell script to implement string operations.
5.	Write a shell script to initialize two numeric variables. Then perform an addition operation on both values and store results in the third variable.
6.	Write a shell script, which takes input from the user at run time. Then calculate the multiplication of given numbers and store to a variable and show the results.
7.	Write a shell script to compute LCM of two numbers
8.	Write a script to implement shell loops.
9.	Write an Interactive Shell Script.
10.	Write a shell script to change date format. Show the time taken in execution of this script.

11.	Write a shell script to change date format. Show the time taken in execution of this script.
12.	Write a shell script to find whether a given number is prime or not.
13.	Write a shell script to print date in different formats.
	Total hours: 30 periods

SEMESTER II

COURSE			Credit Structure & Total Hours					Evaluation Scheme		
S No	**Course Code**	**Course Title**	**L**	**T**	**P**	**C**	**Hrs. / Week**	**IA**	**ET**	**Tot.**
1	24BTPCCX21	Python Programming	3	0	1	4	5	50	50	100
2		Psychology	1	0	1	2	3	50	50	100
3	24BTPCCX22	Data Structures	3	0	1	4	5	50	50	100
4	24BTBSCX21	Engineering Mathematics -II	3	1	0	4	4	50	50	100
5	24BTMCCX21	Indian Knowledge System (SWAYAM -MOOC)	2	0	0	-	2	50	50	100
6	24BTESCX21	Engineering Design and Workshop Practice Simulation	2	0	1	3	4	50	50	100
7		Applied AI	1	0	1	2	3	50	50	100
8	24BTVACX21	Red Hat Academy Course (GSS-RHA)	2	0	0	-	2	50	50	100
		Total	17	1	5	**19**	**28**			

Course code	Course Name	Hours/week			Credit	Max. Marks
		L	T	P	C	
24BTPCCX21	Python Programming	3	0	1	4	100
Pre-requisite	NA					

Evaluation Scheme	Theory	Hours	Marks
	End Semester Examination	2-2.5	50
	Internal Assessment	-	50
	1. Mid Semester Examination (Theory)/Continuous Assessment (Practical)	1.5	30
	2. Attendance	-	10
	3. Assignment/Seminar/Activity/ School Approved Certificate Course/Events/Professional Societies Membership and Participants	-	10

UNIT-I	INTRODUCTION	9
Introduction to Python programming - History of Python - Features of Python - Simple Program in Python - Quotations in Python - Lines and Indentation - Multi-Line Statements - Input Operations - Output Operations.		
UNIT-II	DATA, EXPRESSIONS, STATEMENTS	9
Python interpreter and interactive mode; values and types: int, float, Boolean, string, and list; variables, expressions, statements, tuple assignment, precedence of operators, comments; Illustrative programs: exchange the values of two variables, circulate the values of n variables, distance between two points.		
UNIT-III	CONTROL FLOW STATEMENTS AND STRINGS	9
Control statements- if statement - if-else statement - if-elif-else statement - Nested if - While loop - for loop - else statement used with loops - break statement - continue - pass statement - Strings: string slices - functions and methods - In-built string methods - string formatting operations.		
UNIT-IV	FUNCTIONS AND FILES	9
Functions: return values - parameters - local and global scope - function composition - recursion; Files: Reading and Writing-Format Operators-Filenames and paths.		
UNIT-V	LIST, SET, TUPLES, DICTIONARIES	9
Lists as arrays - Lists: list operations - list slices -list methods - list loop - mutability - aliasing cloning lists - list parameters; Set; Tuples: tuple assignment, tuple as return value; Dictionaries: operations and methods. Application based programming Modules & packages. Introduction to NumPy, pandas, SciPy and Matplotlib Real time Applications.		
	Total hours	45 Periods

Course Outcomes: At the end of the course, students will be able to:

COs	Statements	Bloom's Level
CO1	Understand simple python programs using input output operations.	L2
CO2	Implement python programs using expressions and statements.	L3
CO3	Implement python programs using control flow statements and strings.	L3
CO4	Apply the concepts of functions and files in python programming.	L3
CO5	Analyse and implement various tools, modules and packages for python and develop applications using list, sets, tuples and dictionaries in python.	L4

TEXT BOOK:

1. Martin C. Brown, "The Complete Reference Python" McGraw Hill.,2018.
2. Guido van Rossum and Fred L. Drake Jr, "An Introduction to Python" Revised and updated for Python 3.2, Network Theory Ltd., 2014.

REFERENCES:

3. Charles Dierbach, "Introduction to Computer Science using Python: A Computational Problem-solving Focus", Wiley India Edition, 2015.

4.	John V Guttag, "Introduction to Computation and Programming Using Python", Revised and expandedEdition, MIT Press, 2017.

Course Outcomes with Program Outcomes

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11	PO 12
CO 1	3	3	3	3	2	1	-	-	2	-	-	-
CO 2	2	1	2	3	2	2	-	-	-	-	-	-
CO 3	2	3	2	2	2	2	-	-	-	-	-	-
CO 4	3	3	3	3	2	2	-	-	-	-	-	-
CO 5	2	3	2	2	2	2	1	-	-	-	-	-

S.NO	LIST OF EXPERIMENTS
CO1	Write a program to compute distance between two points taking input from the user
	Write a program (a) For checking whether the given number is even or not (b) To check whether given number is Armstrong or not.
	Write a program using a while loop that asks the user for a number and prints a countdown from that number to zero.
	Find the sum of all the primes below two million. By considering the terms in the Fibonacci sequence whose values do not exceed four million, write a program to find the sum of even valued terms.
CO2	Write a program to count the number of characters in the string and store them in a dictionary data structure.
	Write a program to demonstrate list and tuple in python. Write a program using a for loop that loops over a sequence.
	With a given tuple (1,2,3,4,5,6,7,8,9,10), write a program to print the first half values in one line and the last half values in one line.
	Write a function dups to find all duplicates in the list and a function –unique to find all unique elements of the list.
	Write a function reverse to reverse a list without using reverse function. Also write program to perform addition and multiplication of two square matrices.
	Write a program to implement merge sort, selection sort and insertion sort.
CO3	Using Regular Expressions, develop a Python program to Identify a word with a sequence of one upper case letter followed by lower case letters. b) Find all the patterns of "1(0+)1" in a given string.
CO4	Write a simple script that serves a simple HTTP Response and a simple HTML Page.
CO5	Create a menu drive Python program with a dictionary for words and their meanings. Write functions to add a new entry (word: meaning), search for a particular word and retrieve meaning, given meaning find words with the same meaning, remove an entry, display all words sorted alphabetically.
	The celebrity problem is the problem of finding the celebrity among n people. A celebrity is someone who does not know anyone (including themselves) but is known by everyone. Write a Python program to solve the celebrity problem.
	Total Hours : 30 periods

Course code	Course Name	Hours/week			Credit	Max. Marks
		L	T	P	C	
24BTPCCX22	Data Structures	2	0	1	3	100
Pre-requisite	Basic C Programming and logic building skills					
Evaluation Scheme	**Theory**				**Hours**	**Marks**
	End Semester Examination				2	50
	Internal Assessment				-	50
	1. Mid Semester Examination (Theory)/Continuous Assessment (Practical)				1.5	30
	2. Attendance				-	05
	3. Assignment/Seminar/In class competition/ School Approved Certificate Course / Blended Learning				-	15
UNIT-I	**INTRODUCTION TO DATA STRUCTURES**					**5**
Introduction to Data Structures and their types-Linear & Non Linear Data Structures, Data types – primitive and non-primitive, Abstract Data Type, Efficient use of memory. Introduction to Algorithms and Procedures- Performance Analysis and Measurement (Time and Space complexity of algorithms-Average, Best and Worst case analysis)-Big O, Omega and Theta notations.						
UNIT-II	**SORTING AND SEARCHING**					**5**
Sorting: Insertion sort, Selection sort, Bubble sort, Quick sort, Merge sort, Heap sort Searching: Linear Search and Binary Search Techniques. Hash map						
UNIT-III	**LINEAR DATA STRUCTURES – LIST- STACKS AND QUEUES**					**8**
Array: Representation of arrays, Applications of arrays, Sparse matrix. **Stack:** Operations On Stacks, Applications of Stacks, Polish Expression, Reverse Polish Expression- Infix, Postfix & Prefix conversions, Recursion-Tower of Hanoi. **Queue:** Representation Of Queue, Circular Queue, Priority Queue, Double Ended Queue, Applications of Queue. **Linked List:** Singly Linked List, Doubly Linked list, Circular linked list , Applications of linked list.						
UNIT-IV	**NON-LINEAR DATA STRUCTURES – TREES AND GRAPHS**					**7**
Trees: Basic terminology, Binary Trees, Binary Tree Traversal (Inorder, Postorder, Preorder), Representations of Binary Tree, Application of Trees, Decision tree, Binary Search Tree, AVL tree, B-tree, B+ Tree, Heap, Applications of heap. **Graphs:** Basic terminology and Representations, Graph Traversals- Breadth First Search and Depth First Search, Shortest Path- Dijkstra's algorithm, Prim's and Kruskal's Algorithm for Minimal Spanning tree.						
UNIT-V	**GREEDY METHOD AND DYNAMIC PROGRAMMING**					**5**
Greedy Method Strategy: Optimistic Storage on Tapes, Knapsack Problem Dynamic Programming: Travelling Salesman Problem, Backtracking Strategy- 8-Queens Problem						
	Total hours				**30 Periods**	

Course Outcome(s)	At the end of the course, the students will be able to:	
COs	**Statements**	**Bloom's Level**
CO1	Understand the basics of data structures and data types.	**L2**
CO2	Analyse various applications of linear data structures such as arrays, stacks, queues and linked lists.	**L4**
CO3	Implement different structures like tree, heap and graphs along with their basic operations.	**L3**

CO4	Implement Greedy and Dynamic Programming methods for problem solving.	**L3**
CO5	Differentiate between different techniques of sorting and searching for solving problems.	**L3**

TEXT BOOK:	
1.	Mark Allen Weiss- Data Structures and Algorithm Analysis in C- 2nd Edition- Pearson Education-2016.
2.	Reema Thareja- Data Structures Using C- Second Edition- Oxford University Press- 2011
3.	Data Structures by Forouzan
4.	Data Structures using C and C++ by Langsam, Augestern, Tanenbaum.
REFERENCES:	
1.	Richard F. Gilberg- and Behrouz A. Forouzan- Data Structures - A Pseudocode Approach with C-Thomson 2011.
2.	Aho- J.E.Hopcroft and J.D.Ullman- Data Structures and Algorithms- Pearson education- Asia-2010.

Course Outcomes with Program Outcomes

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11	PO 12
CO 1	3	1	2	3	2	-	-	-	1	-	-	3
CO 2	3	3	2	3	2	-	-	-	1	-	-	3
CO 3	3	3	2	3	2	-	-	-	1	-	-	3
CO 4	3	1	2	3	2	-	-	-	1	-	-	3
CO 5	3	3	2	3	2	-	-	-	1	-	-	3

Practical List: **30 periods**

1. Write a program showing the usage of pointers.
2. Write a program to implement Dynamic Memory Allocation using functions malloc(), calloc(),free() etc.
3. Write a program to implement Tower of Hanoi using 3 discs: (a) with recursion (b) without recursion.
4. Implement a program to perform Stack operations using array with following functions:
   [a] push() [b] pop() [c] peek() [d] isEmpty() [e] isFull() [f] size()
5. Write a program to convert infix notation to postfix notation using stack.
6. Implement a program for Queue that performs following operations using array:
   [a] INSERT [b] DELETE [c] DISPLAY
7. Write a program to implement Circular Queue using array that performs following operations.
   [a] INSERT [b] DELETE [c] DISPLAY
8. Write a menu driven program to implement following operations on the Singly Linked List.
(a) Insert a node at the front of the linked list.
(b) Insert a node at the end of the linked list.

(c) Insert a node such that linked list is in ascending order
(d) Delete first node of the linked list.
(e) Delete a node before specified position.
(f) Delete a node after specified position.

9. Write a program to implement following operations on the doubly linked list.
(a) Insert a node at the front of the linked list.
(b) Insert a node at the end of the linked list.
(c) Delete a last node of the linked list.
(d) Delete a node before specified position.

10. Write a program to implement following operations on a circular linked list.
(a) Insert a node at the end of the linked list.
(b) Insert a node before specified position.
(c) Delete first node of the linked list.
(d) Delete a node after specified position.

11. Write a program to create a binary search tree.
12. Implement inorder, preorder and postorder tree traversals.
13. Write a program to implement Quick Sort.
14. Write a program to implement Merge Sort.
15. Write a program to implement Breadth First Search and Depth First Seach Graph Traversals.
16. Write a program to implement Binary Search.

Course code	Course Name	Hours/week			Credit	Max. Marks
		L	T	P	C	
24BTBSCX21	Engineering Mathematics -II	3	1	0	4	100
Pre-requisite	NA					

Evaluation Scheme	Theory	Hours	Marks
	End Semester Examination	2-2.5	50
	Internal Assessment	-	50
	1. Mid Semester Examination (Theory)/Continuous Assessment (Practical)	1.5	30
	2. Attendance	-	10
	3. Assignment/Seminar/Activity/ School Approved Certificate Course/Events/Professional Societies Membership and Participants	-	10

UNIT-I	SEQUENCES, MATHEMATICAL INDUCTION, AND RECURSION	9

Limit of a sequence, monotone and Cauchy sequences and properties of convergent sequences, examples. Infinite series, positive series, tests for convergence and divergence, integral test, alternating series, Leibnitz test. Mathematical Induction, Strong Mathematical Induction and the Well-Ordering Principle for the Integers, Correctness of algorithms, defining sequences recursively, solving recurrence relations by iteration, Second order linear homogenous recurrence relations with constant coefficients. General recursive definitions and structural induction. Functions Defined on General Sets, One-to-One and Onto, Inverse Functions, Composition of Functions, Cardinality with Applications to Computability.

UNIT-II	FUNCTIONAL SERIES AND MATH FOUNDATION	9

Pointwise and uniform convergence, basic aspects of Power series, Fourier series - Statements, Quantifiers, Operation on sets and functions, Relations, Proofs.

UNIT-III	NUMBER SYSTEM	9

Countability of algebraic numbers, Transcendental numbers and construction of Liouville's number, Equivalence classes, construction of real numbers (using Cauchy sequences), Fermat's little theorem and using it for Miller-Rabin primality test, Wilson's theorem and Primitive root theorem.

UNIT-IV	PROBABILITY	9

Sample space and events, definitions of probability, properties of probability, conditional probability. Random variables: distribution functions, discrete and continuous random variables, moments of random variables, conditional expectation, Chebyshev inequality, functions of random variables. Special Distributions: Bernoulli, Binomial, Geometric, Pascal, Poisson, Exponential, Uniform, Normal distributions, Limit Theorems: Law of large numbers

UNIT-V	STATISTICS	9

Fundamentals of Data: Collection, Summarization, and Visualization; Sampling and Sampling Distributions, Central Limit Theorem; Methods of Estimation, Unbiased estimators; Confidence Interval Estimation: Z-interval, t-interval; Hypothesis Testing, Types of Errors, Rejection Region Approach and p-value Approach.

Total hours	45 periods

Course Outcomes: After completion of course, students would be able to:

COs	Statements	Bloom's Level

CO1	Understand the behaviour of series and their applications.	**L2**
CO2	Explain the various fundamental concepts of Fourier Series.	**L2**
CO3	Understand number system and its applications.	**L3**
CO4	Understand the concept of probability and apply in real life.	**L3**
CO5	Understand and apply the concept of statistics.	**L3**

TEXT BOOK/ REFERENCES:

1. Probability and statistics for Engineers and Scientists, Walpole, Myers, Myers and Ye, Pearson Education, 2012
2. Advanced Engineering Mathematics, Reena Garg, Khanna Book Publishing Co., Delhi
3. Advanced Engineering Mathematics, Wylie and Barrett, McGraw Hill, 1995
4. Advanced Engineering Mathematics, M.D. Greenberg, Pearson Education Asia, 2002
5. Probability and statistics for Engineers and Scientists, Walpole, Myers, Myers and Ye, Pearson Education, 2012

MOOC REFERENCES

1. https://www.udemy.com/course/mathematical-foundation-for-machine-learning-and-ai/
2. Engineering Mathematics – I, Prof. Jitendra Kumar, IIT Kharagpur – NPTEL/SWAYAM
3. Probability and Statistics, Prof. Somesh Kumar, IIT Kharagpur – NPTEL/SWAYAM

Course Outcomes with Program Outcomes

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	1	-	-	-	-	-	-	-	-
CO2	3	3	2	3	-	-	-	-	-	-	-	-
CO3	3	3	3	2	-	-	-	-	-	-	-	-
CO4	3	3	3	3	-	-	-	-	-	-	-	-
CO5	3	3	3	2	1	-	-	-	-	-	-	-

Course code	Course Name	Hours/week			Credit	Max. Marks
		L	T	P	C	
24BTMCCX21	Indian Knowledge system (MOOC)	2	0	0	-	100
Pre-requisite	NA					
Evaluation Scheme	MOOC				Hours	Marks
	Pass/Fail Audit Course - Credit Transfer. No Internal assessment / No End Sem Exam. Certificate of Completion and Final Quiz/Viva.				-	100

UNIT-I	**Introduction & The Vedic Corpus**	**6**

Introduction to IKS, why do we need IKS, Organization of IKS, Historicity of IKS, salient aspects of IKS. Introduction to Vedas, A synopsis of the four Vedas, Sub-classification of Vedas, Introduction to Vedāṅgas, Prologue on Śikṣā and Vyākaraṇa, Basics of Nirukta and Chandas, Kalpa and Jyotiṣa. Distinctive Features of Vedic Life. Introduction to Purāṇa, Purāṇic repository, Introduction to Itihāsas, Key messages in Itihāsas, Wisdom through Nīti-śāstras, Wisdom through Subhāṣita .

UNIT-II	**Number Systems, Units of Measurement & Mathematics**	**6**

Historical evidence of Number systems in India, Salient aspects of Indian Mathematics, Bhūta-Saṃkhyā system, Kaṭapayādi system, Measurements for time, distance, and weight, Piṅgala and the binary system. Introduction to Indian Mathematics, Unique aspects of Indian Mathematics, Indian Mathematicians and their Contributions, Algebra, Geometry, Trigonometry, Binary mathematics and combinatorial problems in Chandaḥ Śāstra, Magic squares.

UNIT-III	**Astronomy, Knowledge Framework and classifications**	**6**

Astronomy - Introduction to Indian astronomy, Indian contributions in astronomy, The celestial coordinate system, Elements of the Indian calendar, Notion of years and months, Pañcāṅga – The Indian calendar system, Astronomical Instruments (Yantras), Jantar Mantar of Rājā Jai Singh Sawai.
Knowledge Framework and classifications: Indian scheme of knowledge, The knowledge triangle, Prameya – A vaiśeṣikan approach to physical reality, Dravyas – the constituents of the physical reality, Attributes – the properties of substances and Action – the driver of conjunction and disjunction, sāmānya, viśēṣa, samavāya, Pramāṇa – the means of valid knowledge, Saṃśaya – ambiguities in existing knowledge, Framework for establishing valid knowledge, Deductive or inductive logic framework, Potential fallacies in the reasoning process, Siddhānta: established tenets in a field of study

UNIT-IV	**Linguistics, Health Wellness and Psychology**	**6**

Introduction to Linguistics, Aṣṭādhyāyī, Phonetics, Word generation, Computational aspects, Mnemonics, Recursive operations, Rule based operations, Sentence formation, Verbs and prefixes, Role of Sanskrit in natural language processing.
Health Wellness and Psychology: Health and wellness in Indian system, Āyurveda: approach to health, Āyurveda: definition of health, Tri-doṣas, Role of agni in health, Sapta-dhātavaḥ: seven-tissues, Psychological aspects of health, Disease management elements, Dinacaryā: daily regimen for health & wellness, Importance of sleep, Food intake methods and drugs, Approach to lead a healthy life, Indian approach to psychology, Basic tenets of Indian psychology, The tri guṇa system & holistic picture of the individual, Prakrti and its evolution and nature of individual, The Pañca-kōṣa framework, Four states of Consciousness, Consciousness studies

UNIT-V	**Town Planning and Architecture**	**6**

Arthaśāstra, Vāstu-śāstra, Eight limbs of Vāstu, Town planning, Temples in India: Stone architecture, Temple architecture in India, Iconography.

Total hours	**30 Periods**

Course Outcome(s)	At the end of the course, the students will be able to:

COs	Statements	Bloom's Level
CO1	To understand Indian Knowledge system concepts & The Vedic Corpus.	**L1**
CO2	To identify and apply the knowledge of Indian Number Systems, Units of Measurement & Indian Mathematics	**L1**
CO3	To understand Astronomy and the Knowledge Framework classifications.	**L2**
CO4	To adapt the learnings from Linguistics, Health Wellness and Psychology for a better life.	**L3**
CO5	To recognize and understand the science behind the Town Planning and Architecture of Indian Temples and other monuments.	**L2**

TEXT BOOK:

1. Mahadevan, B., Bhat Vinayak Rajat, Nagendra Pavana R.N. (2022), "Introduction to Indian Knowledge System: Concepts and Applications", PHI Learning Private Ltd. Delhi.

REFERENCES:

2. Pride of India: A Glimpse into India's Scientific Heritage, Samskrita Bharati, New Delhi.
3. Sampad and Vijay (2011). "The Wonder that is Sanskrit", Sri Aurobindo Society, Puducherry.
4. Acarya, P.K. (1996). Indian Architecture, Munshiram Manoharlal Publishers, New Delhi.
5. Banerjea, P. (1916). Public Administration in Ancient India, Macmillan, London.
6. Kapoor Kapil, Singh Avadhesh (2021). "Indian Knowledge Systems Vol – I & II", Indian Institute of Advanced Study, Shimla, H.P.

MOOC

1. Indian Knowledge System (IKS) : Concepts and Applications in Science
By Prof. B. Mahadevan, Dr. Vinayak Rajat Bhat, Dr. R Venkata Raghavan | Indian Institute of Management Bangalore (IIMB), Chanakya University, Bangalore
https://onlinecourses.swayam2.ac.in/imb24_mg21/preview
2. Indian Knowledge System(IKS): Concepts and Applications in Engineering
By Prof. B. Mahadevan, Dr. Vinayak Rajat Bhat, Dr. R Venkata Raghavan | Indian Institute of Management Bangalore (IIMB), Chanakya University, Bangalore
https://onlinecourses.swayam2.ac.in/imb23_mg53/preview
3. Indian Knowledge System (IKS) : Humanities and Social Sciences
By Prof. B. Mahadevan, Dr. Vinayak Rajat Bhat, Dr. R Venkata Raghavan | Indian Institute of Management Bangalore (IIMB), Chanakya University, Bangalore
https://onlinecourses.swayam2.ac.in/imb24_mg22/preview

Course Outcomes with Program Outcomes

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11	PO 12
CO 1	1	1	-	-	-	1	-	-	1	2	1	1
CO 2	1	1	-	-	-	1	-	-	1	2	1	1
CO 3	2	2	-	-	-	3	-	-	1	2	1	1
CO 4	3	-	-	-	-	3	-	-	1	2	1	1
CO 5	1	-	-	-	-	2	-	-	1	2	1	1

Course code	Course Name	Hours/week			Credit	Max. Marks
	Applied AI	**L**	**T**	**P**	**C**	**100**
		1	0	1	2	
Pre-Requisite	**NA**					
Evaluation Scheme	**Theory**				**Hours**	**Marks**
	Term End Summative Assessment: Case Studies/ PBL/Practical				-	50
	Internal Assessment				-	50
	4) Mid Semester Examination (Theory)/Continuous Assessment (Practical)				1.5	30
	5) Attendance				-	10
	6) Assignment/Seminar/Activity/ School Approved Certificate Course/Events/Professional Societies Membership and Participants				-	10

UNIT-I	**INTRODUCTION TO APPLIED AI**	**3**

Review of key AI concepts from Fundamentals of AI. Exploring different application domains for AI (Law, Management, IT, Design, Dentistry, Arts & Humanities).

UNIT-II	**AI FOR SPECIFIC DISCIPLINES**	**3**

AI applications tailored to each discipline:
Law: Legal research automation, eDiscovery tools, AI-powered contract analysis.
Management: Customer relationship management (CRM) with AI, intelligent marketing strategies, AI-powered decision-making.
IT: Network security with AI, IT infrastructure management using AI, chatbots for customer service.
Design: Personalized design experiences, AI-powered creativity tools, generative design techniques.
Dentistry: Medical imaging analysis with AI, robotic dentistry, AI-assisted treatment planning.
Arts & Humanities: Algorithmic music generation, sentiment analysis of literature, AI for artistic expression.
Case studies.

UNIT-III	**HANDS-ON WITH AI TOOLS**	**3**

Introduction to user-friendly AI platforms and pre-trained models. Hands-on activities: use AI platforms or APIs to perform tasks relevant to any chosen disciplines (e.g., legal document analysis, customer segmentation for marketing, medical image classification).

UNIT-IV	**THE FUTURE OF APPLIED AI**	**3**

Emerging trends in applied AI across various disciplines. Ethical considerations of AI use in specific applications. Potential impact of AI on the future of chosen fields..

UNIT-V	**PROJECT PRESENTATIONS AND DISCUSSIONS**	**3**

Final projects showcasing potential AI applications in their chosen fields, strengths, limitations, and ethical considerations of proposed solutions.

Total hours	**15** periods

Course Outcomes: At the end of the course, the students will be able to:

COs	Statements	Bloom's Level
CO1	Recognize opportunities for applying AI solutions in various fields	**L1**
CO2	Critically assess the effectiveness, limitations, and potential biases of deployed AI systems	**L3**
CO3	Gain hands-on experience using user-friendly platforms and pre-trained AI models.	**L3**
CO4	Explain AI concepts and solutions to a non-technical audience, highlighting benefits and drawbacks.	**L5**
CO5	Explore ethical considerations surrounding AI development and deployment.	**L2**

RESOURCES:	
1.	IBM AI for IT: IBM AI for IT (Solutions and resources for AI in IT infrastructure)
2.	Google AI - Creativity & Design: Google AI Blog (Blog posts on AI applications in design)
3.	Google Cloud AI Platform: Google Cloud AI Platform (Tutorials and documentation for building and deploying AI solutions)

Note : Practical list will be provided by the subject handling faculty.

Course code	Course Name	Hours/week			Credit	Max. Marks
		L	T	P	C	
24BTESCX21	Engineering Design and Workshop Practice Simulation	2	0	1	3	100
Pre-Requisite	NA					

Evaluation Scheme	Theory	Hours	Marks
	End Semester Examination + Final Project	2-2.5	50
	Internal Assessment	-	50
	1. Mid Semester Examination (Practical)/Continuous Assessment (Practical)/ Project Presentation	1.5	30
	2. Attendance	-	10
	3. Assignment/Seminar/Activity/ School Approved Certificate Course/Events/Professional Societies Membership and Participants	-	10

UNIT-I	PROJECTION OF POINTS, LINES, PROJECTION OF PLAIN SURFACES & ORTHOGRAPHIC PROJECTION USING SOLIDWORKS SOFTWARE	6

Introduction to Solidworks Software, Principle Plane of Projection, Orthographic Projection-First angle and Third angle projection, Basic Engineering Drawing Reading methodology, Draw orthographic projection of Simple & Complex objects using Solidworks Software

UNIT-II	2D SKETCH DESIGN USING SOLIDWORKS SOFTWARE	6

Interfaces of Solidwroks Software, Creating Different type of 2D Drawing using Sketch Commands - Line, Circle, Arc, Rectangle, Slot, Polygon, Fillet, Chamfer, Offset, Mirror, Trim, Extend, Linear pattern, Circular pattern, Move, Copy in Solidworks Software.

UNIT-III	3D SOLID MODEL DESIGN USING SOLIDWORKS SOFTWARE	6

Creating Basic & Advanced 3D Models using Feature Commands Extrude feature, Contours, Revolve Features, Swept Features, Reference geometry, Curves, Fillet, Inserting Hole, Chamfer, Rib, 3D Patterns, Draft, mirror.

UNIT-IV	3D ASSEMBLY AND DETAIL DIMENTIONING DRAWING SHEET USING SOLIDWORKS SOFTWARE	6

Introduction to 3D Assembly Modeling & Approaches – Top down and Bottom-up Approach, Creating Assembly Applying Standard Mates- Coincident, Parallel, Perpendicular, Tangent, Concentric, Lock, Distance, Angle. Applying Advanced Mates.

Generating Drawing sheet Using Generating Model View, Projected Views, Inserting Standard 3 View. View creation relative to model, Drawing properties, Creating Dimensions – Annotations, Inserting Annotations, Bill Of Materials, Sheets And Templates, Sheet Formatting in Solidworks Software.

UNIT-V	ADDITIVE MANUFACTURING -3D PRINTING AND SLICING SOFTWARE	6

Introduction To Additive Manufacturing - 3D Printing, Additive Process VS Conventional Process, Introduction Of 3D Printing & Types Of 3D Printer, Introduction to Slicing & 'CURA' Slicing Software, Introduction of Various Printing Methodology, Various Printing Settings, Introduction of Processor, Input filament, Bed Leveling, Adhesion, Basic Part Printing – 1, Basic Part Printing – 2 on 3D Printing Machine.	
Total hours	**30**

Course Outcomes: At the end of the course, the students will be able to:

COs	Statements	Bloom's Level
CO1	Sketch Diagrams with proper Dimensions on Drawings.	**L4**
CO2	Familiar with Office Practice and Engineering Graphic Standards.	**L2**
CO3	Exposure to Solid Modelling & Assembly.	**L4**
CO4	Perform Complex Modeling & also Various Advanced Assemblies in Solidworks	**L3**
CO5	Apply Additive Manufacturing Process & also able to Create & Print Parts with 3D Printing.	**L6**

S.NO	LIST OF PRACTICAL EXPERIMENT
1.	Learning Engineering Drawing Reading and Understanding
2.	Different Types of 2D Sketch Making In Solidworks
3.	2D Sketch and 3D Model Design of Simple and Complex Parts
4.	2D Sketch and 3D Model Design of Card and Pen Holder
5.	2D Sketch and 3D Model Design of Different parts of Drone
6.	3D Model Designs of Machine Parts
7.	3D Assembly (Wheel Support Assembly, Fixture Assembly, Plummer Block Assembly)
8.	3D Assembly (Pneumatic Motor, Drone and Any vehicle like Tractor, Cycle, JCB)
9.	In Detail Manufacturing Drawing sheets of Machine Parts
10.	In Detail Manufacturing Drawing sheets of wheel Support Assembly
11.	Design and Working Simulation of Different Linkage Mechanism
12.	Design and Working Simulation of Robot And Gripper Assembly
13.	Design and Working Simulation of Marble Game, Belt Conveyor
14.	Solidworks Designing of any Live Working Part Modeling & 3D Printing
15.	Wheel Support Assembly, Linkage Mechanism 3D Printing
	Total hours: **30** periods
SOFTWARE	SOLIDWORKS
REFERENCES	Dassault Systems Website, Youtube

Course Outcomes with Program Outcomes

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO 1	3	3	1	2	1	-	-	1	-	1	-	1
CO 2	3	3	3	3	2	-	1	2	1	2	1	1
CO 3	3	3	3	2	3	1	2	2	2	2	-	2
CO 4	3	3	3	2	3	1	2	2	2	2	-	2
CO 5	3	3	3	3	3	2	2	1	2	1	2	2

Course code	Course Name	Hours/week			Credit	Max. Marks
		L	T	P	C	
24BTVACX21	Red Hat System Administration I (RH124) 9.0	2	0	0	-	100
Pre-Requisite	NA					
Evaluation Scheme	**Theory**				**Hours**	**Marks**
	External (End Semester Exam)				2.0	50
	Internal (1) Assignment/Seminar + Quiz-25 Marks / (2) Attendance -5 Marks)				1.5	50

UNIT-I	**Introduction**	**5**

What Is Linux? Access the Command Line, Access the Command Line with the Desktop, Execute Commands with the Bash Shell, Manage Files from the Command Line, Describe Linux File System Hierarchy Concepts, Specify Files by Name, Make Links Between Files, Match File Names with Shell Expansions,

UNIT-II	**Red Hat Enterprise Linux**	**6**

User commands, System calls, Library functions, Special files, File formats, Games and screensavers, Conventions, standards, and miscellaneous, System administration and privileged commands, Linux kernel API, Create, View, and Edit Text Files: Redirect Output to a File or Program, Edit Text Files from the Shell Prompt, Change the Shell Environment, Manage Local Users and Groups: Describe User and Group Concepts, Gain Superuser Access, Manage Local User Accounts, Manage Local Group Accounts, Manage User Passwords

UNIT-III	**Control Access, Monitor and Manage Linux Processes**	**6**

Control Access to Files: Interpret Linux File System Permissions, Manage File System Permissions from the Command Line, Manage Default Permissions and File Access, Monitor and Manage Linux Processes: Process States and Lifecycle, Control Jobs, Kill Processes, Monitor Process Activity, Control Services and Daemons: Identify Automatically Started System Processes, Control System Services.

UNIT-IV	**Configure and Secure SSH and Manage Networking**	**8**

Access the Remote Command Line with SSH, Configure SSH Key-based Authentication, Customize OpenSSH Service Configuration, **Manage Networking:** Describe Networking Concepts, Validate Network Configuration, Configure Networking from the Command Line, Edit Network Configuration Files, Configure Hostnames and Name Resolution,

Install and Update Software Packages: Register Systems for Red Hat Support, Explain and Investigate RPM Software Packages, Install and Update Software Packages with DNF, Enable DNF Software Repositories

UNIT-V	**Access Linux File Systems and Analyze Servers and Get Support**	**5**

Identify File Systems and Devices, Mount and Unmount File Systems, Locate Files on the System, **Analyze Servers and Get Support:** Analyze and Manage Remote Servers, Create a Diagnostics Report, Detect and Resolve Issues with Red Hat Insights

Total hours	**30** periods

Course Outcomes: At the end of the course, the students will be able to:

COs	Statements	Bloom's Level
CO1	Understand the Linux command line interface, including the Bash shell and file system hierarchy, to effectively navigate and execute commands.	**L2**
CO2	Apply system administration tasks in Red Hat Enterprise Linux, such as managing user accounts, editing text files, and handling system calls and library functions.	**L3**

CO3	Analyze and control access to files and Linux processes, including interpreting file permissions, job control, and managing system services.	**L4**
CO4	Evaluate and configure SSH for secure remote access, manage networking settings, and utilize DNF for installing and updating software packages.	**L5**
CO5	Create and manage Linux file systems, analyze remote servers, and develop diagnostic reports to resolve issues using Red Hat Insights.	**L6**

TEXT BOOK:

1. "Red Hat Enterprise Linux 8: The Complete Reference" by Richard Petersen

REFERENCES:

2. Red Hat Certificate System 9 Administration Guide by RedHat.
3. https://rha.ole.redhat.com/rha/app/courses/rh124-9.0/pages/pr01s02

Course Outcomes with Program Outcomes

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	1	3	-	-	1	2	2	-	3
CO2	3	3	3	2	2	-	-	-	2	2	2	3
CO3	3	3	3	2	-	1	-	-	2	2	2	3
CO4	3	3	3	3	3	-	-	-	2	2	2	3
CO5	3	3	3	3	3	-	-	-	2	2	3	3